

Assignment One: Python Basics

Student

- Name: Wang Zhihan
- Course: Edge computing device programming for AI projects
- File submitted: assignment 1_WangZhihan.py
- Deadline: 22/09/2026 11:59pm

GitHub link

<https://github.com/fuqingjiu930/assignment-one-python-basics>

Description

All three tasks of Assignment 1 are written in one Python file, assignment 1_WangZhihan.py, which is stored in the GitHub repository linked above. The file only uses the Python standard library, so it runs on the Jetson Nano without installing anything extra. Running the file opens a small menu where each task can be started on its own, or all three can be run one after another.

How to run

```
cd ~/Desktop
python3 "assignment 1_WangZhihan.py"
```

A menu then appears with the three tasks. Task 3 needs a graphical desktop; the other two tasks also work in a plain terminal.

Task 1: Simple Calculator

The calculator asks for the first number and the second number, lets the user choose an operator from +, -, * and /, and then displays the result, for example "Result: 8 - 3 = 5". Wrong input does not stop the program: dividing by zero prints "Cannot divide by zero", an operator other than the four allowed ones prints "Invalid operation", and anything that is not a number is asked for again.

Task 2: QA Bot

The bot takes the question with input() and chooses its answer with an if / elif / else chain. It knows the five required keywords:

```
hello    -> Bot: Hello! Nice to meet you.
python   -> Bot: Python is a language, and it is very friendly to beginners.
jetson    -> Bot: Jetson Nano is an AI computer made by NVIDIA.
ai        -> Bot: AI means Artificial Intelligence.
name     -> Bot: My name is Python Bot.
```

Capital letters and punctuation are ignored, so Hello!, PYTHON? and "what is AI" are all understood. A question the bot does not know is answered with "Sorry, I don't understand."

Task 3: Turtle Drawing

The drawing task uses turtle with for loops, changing colours and changing lengths. It can draw a square, a triangle, a five-pointed star, a colourful rotating pattern in six colours, and a spider web with twelve spokes in eight colours. The pictures can be drawn one at a time or all together, and clicking the window returns to the menu.

Testing

The program was tested before it was submitted. The calculator was checked with all four operators, with decimal numbers, with a division by zero and with wrong input. The bot was checked with all five keywords in upper and lower case and with questions it does not know. The turtle drawing was checked by confirming that all five pictures are drawn. The program did not stop with an error in any of these tests.